

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 1C

GreenSip Case Workshop: Mathematical Foundations in Action

Bellerbys
Preparation for University Education

Name _____ Class _____ Date _____

WORKSHOP RULES

Calculator allowed. Choose your method and show your working. Use the supplied conversion factors. Keep unrounded values for later tasks. Final masses and percentages: 1 decimal place unless stated otherwise. Money: follow the rounding rule in the task.

TASK C1A

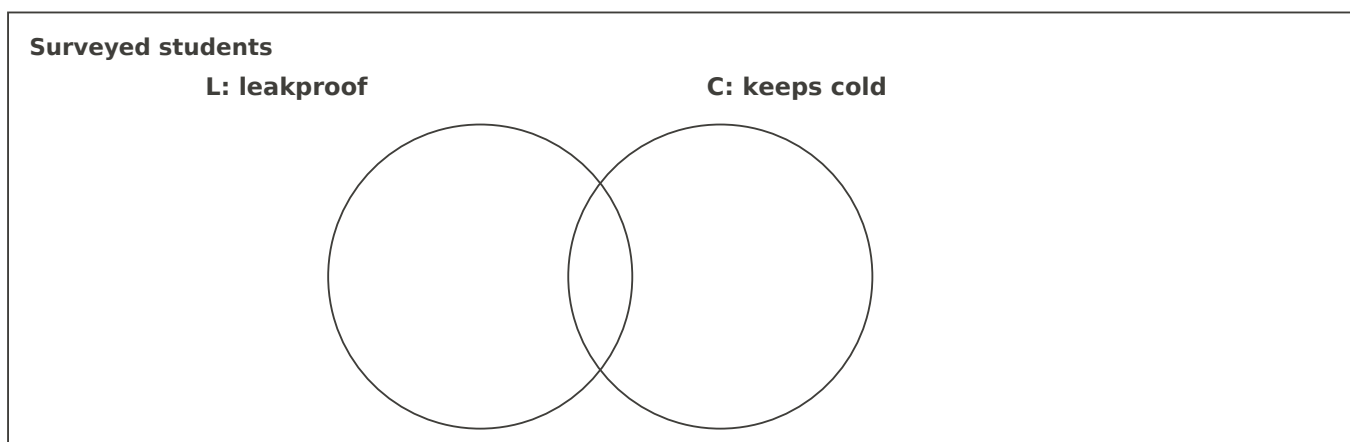
Recover the missing overlap

A survey asked 187 students. Students could select both features.

119 selected leakproof (L), 97 selected keeps cold (C), and 23 selected neither. Each feature total includes the students who selected both.

1. Find the number of students who selected at least one feature.

2. Find how many selected both features. Label all four regions of the Venn diagram.



3. Check that your four regions account for all 187 students.

CUSTOMER EVIDENCE: PERCENTAGES AND CLAIMS

TASK C1B

Same data, different denominator

Use your completed survey diagram from C1A.

1. What percentage of all 187 surveyed students selected at least one feature?

2. Of the students who selected leakproof, what percentage also selected keeps cold?

3. What percentage of all surveyed students selected both? Explain why Q2 and Q3 have different denominators.

THINK

Which statement does the survey support: the result describes these 187 students, or it establishes the percentage for every Bellerbys student? Explain. What would you need to know about how the 187 students were selected?

PRODUCT VERSIONS AND TESTER ROLES**TASK C2A****Versions with restrictions**

A bottle version uses one colour, one lid and one size. There are 5 colours, 4 lids and 3 sizes.

Two distinct colour-lid pairs cannot be made in the largest size. All other versions are available.

1. How many different bottle versions can be made?

2. Share 1,137 bottles equally across these versions. What is the largest whole number per version, and how many bottles are left?

3. THINK: Do these counts show that customers want equal numbers of every version? Explain.

TASK C2B**Testers and different roles**

Choose 4 testers from 11 volunteers. From those 4, appoint one leader and a different person as reporter.

1. How many different groups of 4 testers can you choose?

2. For one chosen group, how many ways can you assign the two jobs?

3. Count all possible results: the group of 4 AND who holds each job. Changing either the group or a job holder gives a different result.

OPTIONAL EXTRA Maya is one of the 11 volunteers. She must be among the 4 testers and cannot be leader. She may be reporter. How many group-and-job results are possible?

MIXED UNITS: BOTTLE MASS AND SHIPMENT**TASK C3A****A mixed-unit packing equation**

Eight identical bottles plus a 0.35 kg carton weigh 5.8 lb. Use $1 \text{ kg} \approx 2.205 \text{ lb}$. Let x be the mass of one bottle in kilograms.

1. Form an equation and find one bottle's mass in grams, to the nearest gram.

2. Check your result against the packed mass. Keep the unrounded x for C3B and C4A.

TASK C3B**The packed shipment**

Ship 1,137 bottles. Use the unrounded bottle mass x from C3A. Each carton holds up to 8 bottles. The carton itself weighs 0.35 kg, regardless of how many bottles it contains.

Mass limit: 60 stone. Use $1 \text{ stone} = 14 \text{ lb}$ and $1 \text{ kg} \approx 2.205 \text{ lb}$.

1. What is the smallest number of cartons needed? Explain your rounding.

2. Find the total mass of the bottles AND all cartons, in pounds.

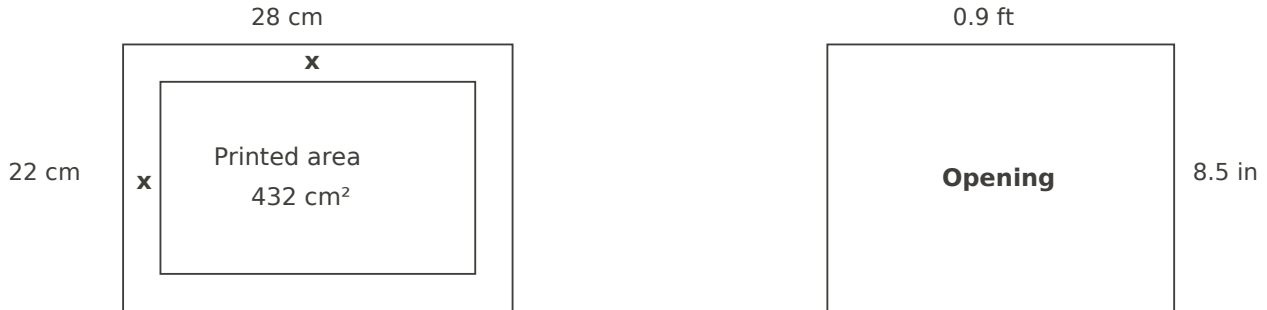
3. Is this below or above the limit? By how many pounds?

OPTIONAL EXTRA With the same packaging rules and mass limit, what is the maximum number of bottles that can be shipped?

ALGEBRA: BORDER WIDTH AND PHYSICAL LIMITS**TASK C3C****The border and the opening**

The whole card is 28 cm by 22 cm. A border of the same width x on all four sides surrounds a printed area of 432 cm^2 .

The whole card must pass flat through an opening 0.9 ft by 8.5 in, with corresponding edges parallel. Use $1 \text{ ft} = 12 \text{ in}$ and $1 \text{ in} = 2.54 \text{ cm}$.



Schematic - not to scale

- Find the border width x in cm. Form an equation, find both roots and explain which root fits the card.

- Find the printed length and width. Check that their area is 432 cm^2 .

- Will the WHOLE card fit without folding? Show the two dimension comparisons.

OPTIONAL EXTRA If the printed area changes to 430 cm^2 , find the border width to 2 decimal places. Choose a suitable method.

PERCENTAGE LOSS AND MIXTURE PLANNING

TASK C4A

Two stages of material loss

GreenSip needs 1,137 bottles that pass inspection. Use the unrounded mass of one bottle from C3A. Cartons are not part of the bottle material.

Manufacturing loses 7.5% of the starting material. Inspection rejects 2% of the mass left after manufacturing.

1. Find the total mass of bottles that pass inspection, in kilograms.

2. How much material must GreenSip start with? How much is lost or rejected in total?

3. Is the overall loss $7.5\% + 2\% = 9.5\%$? Calculate the correct percentage to 2 decimal places and explain.

TASK C4B

A three-material mixture

Split the unrounded input mass from C4A in the ratio A:B:C = 7:5:3. Assume both loss stages affect all three materials proportionally.

Material A comes in full 25 lb sacks. Use $1 \text{ kg} \approx 2.205 \text{ lb}$.

1. Calculate the required mass of A, B and C in kilograms.

2. Find the minimum number of sacks of A to order.

3. How much unused A remains after making the exact mixture? Give your answer in kg.

OPTIONAL EXTRA If you use every kilogram of A ordered, how much additional B and C would keep the ratio 7:5:3?

VOLUME AND PERCENTAGES

TASK C4C

Which gallon?

A wash station holds 8.5 US liquid gallons of cleaning liquid. Before use, 5% spills. Each bottle wash consumes 125 mL. Count complete washes only.

Use 1 US liquid gallon $\approx$ 3.78541 L, 1 Imperial gallon $\approx$ 4.54609 L, and 1 L = 1,000 mL.

1. How many complete bottle washes can the station supply?

2. A colleague reads the label as 8.5 Imperial gallons. How many complete washes would they predict? By how many would they overestimate?

3. Explain why the label '8.5 gallons' is not precise enough.

Conversion factors: NIST SP 811 and Handbook 44 (rounded for these tasks).

TASK C5A

Orders, discounts and rounding

GreenSip has 660 confirmed one-bottle orders. Student:staff = 7:4. List price: ¥137.50. Student discount: 17.5%. Staff discount: 8%.

Assume all orders are paid. Round EACH final selling price to 2 decimal places before multiplying by the number of orders.

1. Find the number of student and staff orders and each charged price.

2. Find the total money received. Is it above or below ¥78,000? Find the difference and percentage above/below this target (3 decimal places).

3. Recalculate using unrounded prices until the final total. How different is the answer? Which method follows the charging rule?

TWO DISCOUNTS AND INDEPENDENT EXIT CHECK**TASK C5B****Two discounts on one bottle**

Original price: ¥137.50. First discount: 17.5%. Price after rounding: ¥113.44. A second discount reduces ¥113.44 to a final price of ¥99.83.

Use these displayed prices. Give both percentage answers to 2 decimal places.

1. What is the second discount percentage, r ? This is the reduction from ¥113.44 to ¥99.83.

2. What is the total percentage reduction from ¥137.50 to ¥99.83?

3. Why can we not add 17.5% and r % to answer Q2?

THINK

Use C5A. Does that total apply to the 660 paid orders or every future batch? Does it prove all 1,137 bottles will sell? Explain and name one useful piece of additional evidence.

Work independently. Choose your methods and include units.

TASK C6**Individual exit check**

EcoCup needs 417.3 kg of accepted material. Production loses 9% of its input mass. The input ratio is A:B:C = 5:3:2. Loss affects the materials proportionally.

The supplier offers 30 sacks, each containing 35 lb of input mixture. Use $1 \text{ kg} \approx 2.205 \text{ lb}$.

1. Find the required input mass and the mass of each material.

2. Are 30 sacks enough? Show the surplus or shortage in kilograms.

3. State one assumption you would check before using these calculations as a production plan.
